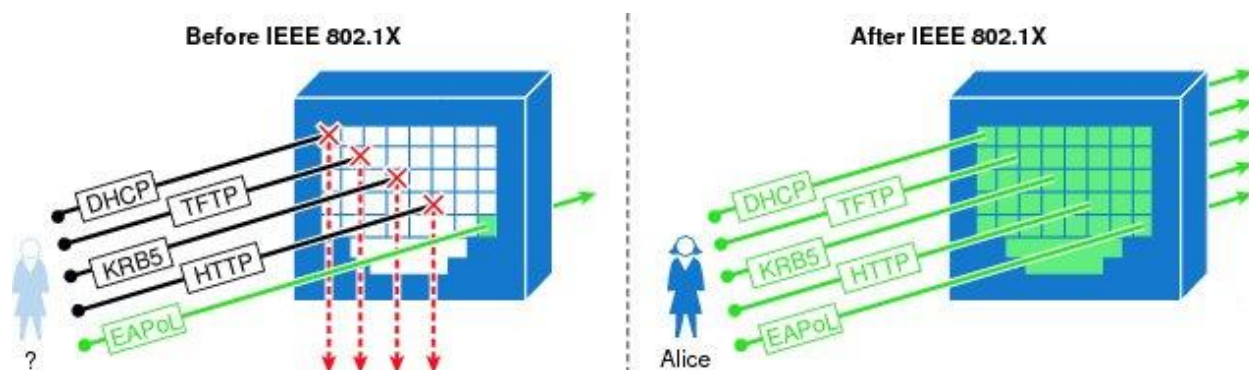


Implement 802.1X:

- o IEEE 802.1x (Dot1x) is a standard set by the IEEE 802.1 working group Organization.
- o IEEE 802.1X (Dot1x) commonly referred or called Dot1x Port Based Authentication.
- o IEEE 802.1X (Dot1x) authentication is an OSI Model Data Link Layer (Layer 2) protocol.
- o IEEE 802.1X (Dot1x) provide port-based network access control using authentication.
- o IEEE 802.1X (Dot1x) authentication method service is called port-level authentication.
- o IEEE 802.1X (Dot1x) is defined as a standard for "Port-Based Network Access Control".
- o The protocol used in IEEE 802.1x (Dot1x) is called EAP Encapsulation over LANs (EAPOL).
- o IEEE 802.1x (Dot1x) uses Extensible Authentication Protocol (EAP) to exchange messages.
- o IEEE 802.1x (Dot1x) which referred Dot1x is used mainly for port-based authentication.
- o Dot1x can be used to prevent unauthorized devices from gaining access to network.
- o Dot1x standards provide authentication & authorization services at network port level.
- o IEEE 802.1x (Dot1x) provide real security for wired and wireless networks at layer two.
- o IEEE 802.1x (Dot1x) authentication is a Client and Server based authentication protocol.
- o Before authentication, identity of endpoint is unknown & all traffic is blocked except EAPoL.
- o Once the user credentials are successfully verified, then other user traffic is permitted.



802.1X Components:

- o The IEEE 802.1x framework defines three roles in the authentication process:

Supplicant (Client or Host):

- o Supplicant is the user or device that wants access the wireless or wire network.
- o Supplicant, client or host is the device or user requiring authentication.
- o Supplicant is also known Client, as 802.1x Port-Based Authentication.
- o Supplicant is the Workstation that is connected through Network Access Switch.
- o Supplicant is the workstation request for accessing the network resources.
- o Supplicant is the Workstation or Client must be using 802.1x Client software.
- o Supplicant could be an end-user device, a printer, Fax machine, PC or an IP phone.
- o The supplicant is the 802.1x software that runs on the endpoint or end Device.
- o All Windows Operating System has own common native supplicant for wired networks.

Authentication Server:

- o Authentication Server is a device that processes authentication such as RADIUS.
- o Authentication Server is the device that authenticates the Supplicant or client.
- o The entity that validates the identity of the supplicant and notifies the authenticator.
- o Authentication Server notifies authenticator to allow or deny the client request.
- o For example, RADIUS server, such as ACS, can provide authentication server services.

Authenticator (Switch, AP):

- o Device between supplicant & authentication server that facilitates authentication.
- o The client, user or supplicant is normally directly connected to the authenticator.
- o For example, switch or wireless access point provide authenticator services to clients.
- o Authenticator is the network device that is acting as a “gatekeeper” to the network.
- o Authenticator is typically a Cisco Layer 2 or 3 Switch or Wireless LAN Controller (WLC).

